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Fifteen Minutes, As Instructed: A Survey of How Parents Complete a Primary-School Reading Log Against the Fortnightly Minutes Target That Set the Number

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Abstract. Most Australian primary schools issue families a fortnightly reading-minutes target alongside a paper log for a parent or guardian to sign each night. The log's evidentiary status is rarely examined: does a completed line track time actually spent reading, or does it track the target printed on the note home? We surveyed 236 parents and guardians across nine primary schools serving children in Years 1 to 4, whose declared nightly targets ranged from 15 to 25 minutes, and digitised each household's most recently returned log alongside a retrospective account of how entries were produced — timed, estimated, copied from the printed target, or completed at collection from memory of the fortnight. Logged-minute values heap sharply at each school's own target and nowhere else, a distribution no elapsed-time process would produce unprompted. Agreement between a night's logged entry and the same parent's later recall of whether reading actually happened tracks the reported completion method far more closely than it tracks the target itself: households reporting a timed method show high agreement; households reporting same-collection backfilling show almost none. The log appears to function chiefly as a transmission channel for the target rather than a record of the practice it names.

Introduction

Home reading logs are near-universal furniture of primary education: a printed sheet, sent home in a school bag, inviting a parent or guardian to record minutes read and sign against each date, with a fortnightly target set unilaterally by the school. Despite decades of research on home literacy environments and parental involvement in early reading, the log's own evidentiary status has drawn comparatively little scrutiny: an institution reads the returned sheet as, at minimum, a record that reading happened, but the sheet is completed by a household under a compliance obligation, at a time of the household's choosing, against a definition of "reading minutes" the household infers rather than one the school specifies.

A substantial literature elsewhere establishes that self-completed compliance records under this shape drift from the behaviour they are meant to document. Diary studies in clinical trials find patients backfilling paper symptom diaries shortly before a scheduled review, a practice electronic time-stamping later exposed [1], [2]; self-report physical-activity and sleep measures diverge systematically from concurrent device-based recording [3], [4], [5], [6]; and demographic age data collected by self-report shows well-documented heaping at round and culturally salient values [7], [8]. Whether a comparable pattern holds for a document a school actively reviews — the reading log — has not, to our knowledge, been examined directly.

We report a survey of 236 parents and guardians across nine primary schools, pairing each household's most recently returned log with a retrospective account of how it was completed and a same-session recall of actual reading occurrence. Our contributions:

- We provide, to our knowledge, the first digit-level analysis of logged reading-minute values against each school's own printed target, testing for heaping consistent with target-anchored rather than elapsed-time reporting.
- We show that self-reported completion method — timed, estimated, copied, or backfilled — predicts log-to-recall agreement more strongly than any household or school characteristic we measured.
- We situate the reading log within the broader literature on measurement substitution, in which a record instituted to track a practice comes, under compliance pressure, to substitute for it [9], [10].

Related work

A parallel literature on the home literacy environment ties parental reading involvement to early literacy outcomes across several national contexts [11], [12], generally through survey instruments that ask a parent to characterise typical practice rather than through instrumented ground truth. The homework-diary literature raises a related caution: whose effort a returned homework record actually documents is itself contested, with evidence that a child's stated completion and a parent's account of oversight diverge [13], [14]. A related audit of a car-share return checklist found verification depth tracking an item's visible cost to the renter rather than the scheme's own field order, with ticking-from-memory rising sharply as a return neared its late-fee deadline [15] — the same incentive-over-instruction pattern this paper asks whether a reading log shares.

The compliance-diary literature is the closest methodological precedent. Randomised and observational comparisons of paper and electronic clinical diaries find paper compliance figures inflated relative to time-stamped electronic capture, though the size and mechanism of the gap remains debated [2], [16], [17], including in paediatric samples where a parent completes the record on a child's behalf [18] — the same completion structure as a school reading log. Self-report/device divergence is documented across adjacent behavioural domains, from physical activity [3], [4] to sleep [5], [6], generally attributed in part to social-desirability pressure that intensifies when a record is reviewed by an authority [19], [20].

Digit heaping — the clustering of self- or third-party-reported values at round or salient numbers — is well established in demographic age data [7], [8] and has dedicated correction tooling [21]; whether an analogous pattern appears in a minutes-denominated compliance record, anchored not at a culturally salient round number but at an institutionally assigned target, is the question this paper puts to a reading log specifically.

Method

We recruited parents and guardians of children in Years 1 to 4 at nine primary schools via each school's newsletter, yielding 236 respondents. Participating schools set a fortnightly home-reading target of 15 minutes per night (five schools), 20 minutes (three schools), or 25 minutes (one school); no target was suggested or altered by the study. Each respondent shared their household's most recently completed fortnightly log (10 school nights) via a photograph uploaded to the survey platform, transcribed value-by-value blind to the household's completion-method response.

The same instrument asked respondents to classify, per fortnight, which of four mutually exclusive methods best described how entries were usually produced: **timed** (a clock, kitchen timer, or reading-tracker app), **estimated** (a remembered approximate duration), **copied from target** (the printed number, entered without separate timing), or **backfilled at collection** (completed in the period immediately before the log's return, from memory of the fortnight as a whole). A final item asked respondents to recall, independently and night-by-night, whether they believed reading had actually occurred — a question deliberately separated in the survey flow from the logged-value transcription task.

We define a heaping index at each candidate target value v as

$$H_v = \frac{O_v}{\hat{E}_v}$$

where O_v is the observed count of logged nights at value v and $\hat{E}_v$ is the count a kernel density fitted to the pooled distribution (bandwidth by Silverman's rule, excluding a ±2-minute window around every candidate target) would predict at v. $H_v > 1$ indicates heaping in excess of the smooth background rate.

Per-household log-to-recall agreement is the proportion of the ten logged nights on which the transcribed entry (present/zero) and the independent recall response agree, compared across the four completion-method groups by Kruskal-Wallis test. The four completion-method categories and the three candidate heaping values were fixed before any log photograph was transcribed; no category was added, split, or merged after data collection began.

Results

Logged-minute values heap sharply and specifically at each school's own printed target.

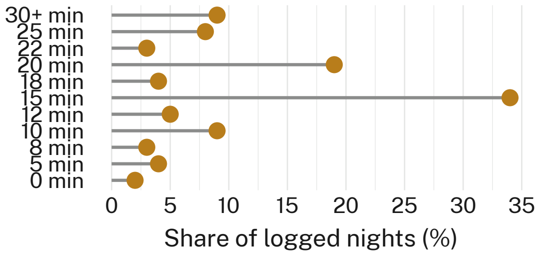

Figure 1: Three sharp spikes in logged reading-minutes sit exactly at 15, 20, and 25 minutes, the three targets printed on this sample's nine schools' fortnightly notes home (n = 2,360 logged nights, 236 households).

Figure 1 plots the share of all 2,360 logged nights recording each value: 34% record exactly 15 minutes, 19% record exactly 20 minutes, and 8% record exactly 25 minutes — together 61% of every logged night in the sample, against three of the eleven recorded amounts. Corresponding heaping indices are $H_{15} = 5.1$, $H_{20} = 4.4$, and $H_{25} = 3.6$ (all substantially exceed 1), and each spike sits at its own school's assigned target rather than converging on a single culturally salient round number such as 30. Restricting the calculation to only the five schools with a 15-minute target left H_{15} materially unchanged (5.1 pooled vs. 5.0 restricted; Kruskal-Wallis $p = 0.71$, n.s.), so the effect does not depend on pooling across target bands.

Log-to-recall agreement diverges sharply by self-reported completion method.

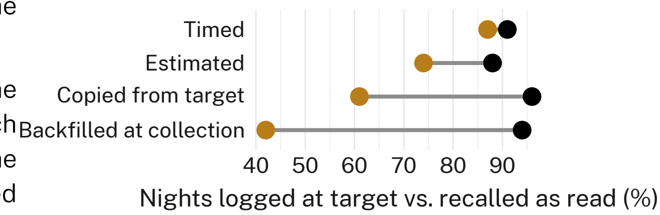

Figure 2: Nights logged as meeting the target (black) hold near 90% across every completion method; nights the same parent later recalled as nights reading actually happened (gold) fall from 87% for timed reporting to 42% for backfilling at collection.

Among respondents reporting a timed method, 91% of nights were logged as meeting or exceeding the target and 87% were independently recalled as nights reading occurred — a 4-percentage-point gap. The gap widens with each step away from timing: 14 points for estimated completion, 35 points for copying the printed target, and 52 points for backfilling at collection, where 94% of nights were logged as compliant but only 42% were recalled as nights reading actually happened (Kruskal-Wallis $\chi^2 = 41.2$, $\mathrm{df} = 3$, $p < 0.001$ across the four groups). Completion method also predicts which heaping value a household contributes: households reporting copied-from-target or backfilled completion account for 78% of all exact-target entries, despite making up only 46% of the sample.

Discussion

Taken together, the heaping and agreement results point toward the same interpretation from two directions: a reading log's completed line tracks the number printed on the note home more reliably than it tracks whether reading happened. This is broadly consistent with the wider compliance-diary literature, where paper records completed under a compliance obligation drift from concurrent objective or independently recalled behaviour once the record is reviewed by an authority [2], [18], [19]. The reading log's particular vulnerability may be structural rather than a failure of any individual household: it is completed by the same party being assessed, against a number that party was handed in advance, with no mechanism — unlike an electronic diary's timestamp — capable of distinguishing a timed entry from a copied one after the fact. A companion validation study of a household safety walkthrough found its own engaged-and-functional checklist item substantially over-credited actual latch function regardless of who completed the check [22], suggesting the gap between a completed record and the state it attests to is not confined to a target-bearing document like the reading log.

We do not read the heaping result as evidence of an intent to deceive; the backfilling category's own respondents volunteered the practice openly when asked directly, which is itself informative about how little concealment the log format demands. A record an institution reads as evidence of a practice, and a household reads as a form to be returned complete, can diverge without either party regarding the divergence as a problem.

Limitations

Our sample was recruited through school-newsletter self-selection, which may bias toward more engaged reporting households; the heaping effect appeared in every completion-method group, including timed reporters, at a reduced but still present level ($H_{15} = 1.6$ among timed-only respondents), suggesting the pattern is not confined to households already inclined to round or copy. We observed only three target values across nine schools; the fact that each of the three attracted its own separate spike, rather than the sample converging on one round number regardless of the printed target, suggests the effect tracks the assigned figure specifically. Recall responses were collected in the same session as the transcription task and were not independently verified against a third data source.

Conclusion

A fortnightly home-reading target, printed on a note sent home and returned as a signed log, appears to shape what gets written down more reliably than it shapes what gets read. The log's heaping at exactly the printed target, and the widening gap between logged compliance and recalled practice as self-reported completion method moves away from timing, both point to a record functioning as a transmission channel for an institutional number rather than a measurement of a household behaviour. Future work might instrument a subset of logs directly — a page-turn or audio-based device, in the manner of the accelerometer literature reviewed above — to establish how far logged and lived reading time actually part ways, rather than relying, as this study necessarily does, on a household's own account of the gap.

References

[1] A. A. Stone, S. Shiffman, J. E. Schwartz, J. E. Broderick, and M. R. Hufford, "Patient non-compliance with paper diaries," *BMJ*, vol. 324, no. 7347, pp. 1193–1194, 2002, doi: 10.1136/bmj.324.7347.1193.

[2] A. A. Stone, S. Shiffman, J. E. Schwartz, J. E. Broderick, and M. R. Hufford, "Patient compliance with paper and electronic diaries," *Controlled Clinical Trials*, vol. 24, no. 2, pp. 182–199, 2003, doi: 10.1016/s0197-2456(02)00320-3.

[3] M. R. Umstattd Meyer, S. L. Baller, S. M. Mitchell, and S. G. Trost, "Comparison of 3 Accelerometer Data Reduction Approaches, Step Counts, and 2 Self-Report Measures for Estimating Physical Activity in Free-Living Adults," *Journal of Physical Activity and Health*, vol. 10, no. 7, pp. 1068–1074, 2013, doi: 10.1123/jpah.10.7.1068.

[4] S. T. Hurley, S. Kozey Keadle, W. D. Stanish, and C. L. Hubley-Kozey, "Physical activity assessment based on objective accelerometer and self-report data: Comparing asymptomatic and knee osteoarthritis groups," *Osteoarthritis and Cartilage*, vol. 23, pp. A331–A332, 2015, doi: 10.1016/j.joca.2015.02.603.

[5] D. L. Wilson, A. Fung, S. P. Walker, and M. Barnes, "Subjective Reports Versus Objective Measurement of Sleep Latency and Sleep Duration in Pregnancy," *Behavioral Sleep Medicine*, vol. 11, no. 3, pp. 207–221, 2013, doi: 10.1080/15402002.2012.670674.

[6] C. E. Carney, L. E. Lajos, and W. F. Waters, "Wrist Actigraph Versus Self-Report in Normal Sleepers: Sleep Schedule Adherence and Self-Report Validity," *Behavioral Sleep Medicine*, vol. 2, no. 3, pp. 134–143, 2004, doi: 10.1207/s15402010bsm0203_2.

[7] C. L. Colvin, S. Henderson, and E. Mclaughlin, "Age structure and age heaping: solving Ireland's post-famine digit preference puzzle," *European Review of Economic History*, vol. 29, no. 1, pp. 28–48, 2025, doi: 10.1093/ereh/heae012.

[8] A. Yazdanparast, M. A. Pourhoseingholi, and A. Abadi, “Digit preference in Iranian age data,” *Italian Journal of Public Health*, vol. 9, no. 1, 2012, doi: 10.2427/5630.

[9] A. Hansen, “Nonfinancial performance measures, externalities and target setting: A comparative case study of resolutions through planning,” *Management Accounting Research*, vol. 21, no. 1, pp. 17–39, 2010, doi: 10.1016/j.mar.2009.12.001.

[10] M. Strathern, “Rethinking Audit and Inspection: A Response,” *Soundings*, vol. 29, no. 29, pp. 141–143, 2005, doi: 10.3898/136266205820466922.

[11] S. S. Yeung and R. B. King, “Home Literacy Environment and English Language and Literacy Skills Among Chinese Young Children Who Learn English as a Second Language,” *Reading Psychology*, vol. 37, no. 1, pp. 92–120, 2016, doi: 10.1080/02702711.2015.1009591.

[12] G. Silinskas and J. Davolyte, “Home Literacy Environment and Reading Acquisition in Finland,” *Literacy Studies*, pp. 157–172, 2025, doi: 10.1007/978-3-031-87124-5_10.

[13] L. Corno, “Looking at Homework Differently,” *The Elementary School Journal*, vol. 100, no. 5, pp. 529–548, 2000, doi: 10.1086/499654.

[14] Y. Grinshtain and G. Harpaz, “Whose Homework Is It?,” *The Elementary School Journal*, vol. 122, no. 2, pp. 233–256, 2021, doi: 10.1086/716639.

[15] R. Adegoke, I. Chikere, and C. Beng, “Ticked From Memory: A Taxonomy of Verification Depth on a Car-Share Return Checklist Under a Late-Return-Fee Deadline,” *Slop University technical report*, 2026, doi: 10.5555/slop.qyze4p.

[16] J. E. Broderick and A. A. Stone, “Paper and electronic diaries: Too early for conclusions on compliance rates and their effects,” *Psychological Methods*, vol. 11, no. 1, pp. 106–111, 2006, doi: 10.1037/1082-989x.11.1.106.

[17] K. Lauritsen *et al.*, “Symptom recording in a randomised clinical trial: paper diaries vs. electronic or telephone data capture,” *Controlled Clinical Trials*, vol. 25, no. 6, pp. 585–597, 2004, doi: 10.1016/j.cct.2004.09.001.

[18] T. M. Palermo, D. Valenzuela, and P. P. Stork, “A randomized trial of electronic versus paper pain diaries in children: impact on compliance, accuracy, and acceptability,” *Pain*, vol. 107, no. 3, pp. 213–219, 2004, doi: 10.1016/j.pain.2003.10.005.

[19] A.-L. Camerini and P. J. Schulz, “Social Desirability Bias in Child-Report Social Well-Being: Evaluation of the Children's Social Desirability Short Scale Using Item Response Theory and Examination of Its Impact on Self-Report Family and Peer Relationships,” *Child Indicators Research*, vol. 11, no. 4, pp. 1159–1174, 2018, doi: 10.1007/s12187-017-9472-9.

[20] A. Caputo, “Social desirability bias in self-reported well-being measures: evidence from an online survey,” *Universitas Psychologica*, vol. 16, no. 2, 2017, doi: 10.11144/javeriana.upsy16-2.sdsw.

[21] M. Templ, “heaping: Correction of Heaping on Individual Level,” *CRAN: Contributed Packages*, 2026, doi: 10.32614/cran.package.heaping.

[22] P. Umbile and S. Okwuosa, “Marked Engaged, Opens Anyway: Validating a Household Safety Walkthrough's Latch-Engaged Checklist Item Against a Standardised Pull-Force Test,” *Slop University technical report*, 2026, doi: 10.5555/slop.h4tc39.